

At the heart of the circuit is microcontroller AT89C51. It is a low-power, high-performance, 8-bit microcontroller with 4kB Flash programmable and erasable read-only memory. It has 128 bytes of RAM, 32 input/output (I/O) lines, two 16-bit timers/counters, a five-vector two-level interrupt architecture, on-chip oscillator and clock circuitry.

The 11.0592MHz crystal provides the basic clock frequency to the microcontroller. Port pin P2.0 of the microcontroller provides the ring tone melody signal for speaker LS1. Transistor BC337 is used for amplification. The power-on reset signal for the microcontroller is generated by the combination of capacitor C3 and resistor R2. Switch S1 provides manual reset to the microcontroller.

The 230V AC mains is stepped down by transformer X1 to deliver the secondary output of 9V, 500 mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4,

filtered by capacitor C1 and regulated by IC 7805 (IC2). Capacitor C2 bypasses the ripples present in the regulated power supply. LED1 acts as the power-on indicator and resistor R1 limits the current through LED1.

An actual-size, single-side PCB for the microcontroller-based ring tone generator is shown in Fig. 4 and its component layout in Fig. 5.

Software

The program plays "happy birthday to you" in RTTTL ring tone format using the microcontroller AT89C51. The source program, written in Assembly language and assembled using assembler ASM51, is self-explanatory and easy to understand.

Initialise timer 0 and timer 1 as 16-bit timers with predetermined value. When you start timer 0, the data pointer register is loaded with memory address labeled as 'SONG.' After playing the current note, the control jumps to the next note and it starts playing.

This process continues until the end of music data is reached. Thereafter, it starts playing the music from the beginning.

Nokia RTTTL ringtones can be downloaded from the following web-sites:

1. http://www.2thumbswap.com/members/tones/nokia/tones_nokia_latest.html
2. <http://nokiadone.ifrance.com/nokiadone/rtttl.htm>
3. <http://ringtones.frostzone.com/index.htm>
4. <http://arcadetones.emuunlim.com/files.htm>

RTTTL ringtones can also be tested on the computer, the software for which can be downloaded from the link 'http://arcadetones.emuunlim.com/files/nokring_full.zip.'

(Refer "A Cell Phone Player Embedded in Microcontroller" by Manoel Gomes de Andrade for details.)

EFY note. The source code and other relevant files of this article are included in this month's EFY-CD.

RINGTONES.ASM

<pre> Smod51 FLAG EQU 07FH SPEAKER EQU P2.0 BEGIN: ORG 0000H AJMP MAIN ISR: ORG 000BH LCALL ACTION RETI ORG 0050H MAIN: MOV TMOD,#00010001B ; MOV IE,#82H ; MOV TH0,#0DCH ; MOV TLO,#00H ; SETB TR0 MOV DPTR,#SONG MOV R0,#02 LOOP1: ACALL PLAY NOTE CJNE R0,#02,CARRY_ON SJMP MAIN ;end of music ;has been detected so start again CARRY_ON: ACALL PLAY SJMP LOOP1 PLAY: CLR EA SETB FLAG SETB EA JNB FLAG,STOP CJNE R0,#0,START SJMP RUN START: MOV T1,R2 MOV T1,R3 SETB T1 HERE: JNB T1,HERE CLR T1 CLR TR1 CPL SPEAKER SJMP RUN STOP: CLR SPEAKER </pre>	<pre> PLAY NOTE: CONTINUE: CLR A MOV A,#A+DPTR MOV R0,A INC DPTR CLR A MOV A,#A+DPTR MOV R1,A INC DPTR CLR A MOV A,#A+DPTR MOV R2,A INC DPTR CLR A MOV A,#A+DPTR MOV R3,A INC DPTR RET ACTION: CLR TR0 CJNE R1,#0,DO2 CLR FLAG SJMP DO3 DO2: DEC R1 DO3: MOV TH0,#0DCH MOV TLO,#00H SETB TR0 RET SONG: ; HAPPY BIRTHDAY ; R0 R1 R2 R3 DB 1, 12, 0FCH, 08FH ;R0=1 indicates there is a note to be played DB 0,6,0,0 ;R0=0 indicates there is no note to be played DB 1,6,0FCH,08FH DB 0,6,0,0 DB 1,24,0FCH,0EFH DB 0,6,0,0 DB 1,24,0FCH,08FH DB 0,6,0,0 </pre>	<pre> DB 1,24,0FCH,06CH DB 0,6,0,0 DB 1,48,0FCH,045H DB 0,12,0,0 DB 1,12,0FCH,08FH DB 0,6,0,0 DB 1,6,0FCH,08FH DB 0,6,0,0 DB 1,24,0FCH,0EFH DB 0,6,0,0 DB 1,24,0FCH,08FH DB 0,6,0,0 DB 1,24,0FCH,0B4H DB 0,6,0,0 DB 1,48,0FCH,06CH DB 0,24,0,0 DB 1,12,0FCH,08FH DB 0,6,0,0 DB 1,6,0FCH,08FH DB 0,6,0,0 DB 1,24,0FCH,048H DB 0,6,0,0 DB 1,24,0FCH,0E9H DB 0,6,0,0 DB 1,24,0FCH,06CH DB 0,6,0,0 DB 1,12,0FCH,023H DB 0,6,0,0 DB 1,6,0FCH,023H DB 0,6,0,0 DB 1,24,0FCH,0E9H DB 0,6,0,0 DB 1,24,0FCH,06CH DB 0,6,0,0 DB 1,24,0FCH,0B4H DB 0,6,0,0 DB 1,48,0FCH,06CH DB 2,0,0,0 ;R0=2 indicates end of music END </pre>
--	--	---